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Every mandatory requirement and agent task

Twelve mandatory requirements (M1-M12), phase one at full occupancy

	Mandatory requirement	This scheme	Evidence	Answered in
M1	Hourly carbon-free energy match	62 %	computed	§3. Energy and compute
M2	Peak IT load in phase one	12 MW	assumed	§3. Energy and compute
M3	Annual water demand, by use	1,000,000 m3/yr	assumed	§4. Water
M4	Cooling water intensity (WUE)	0.02 L/kWh	assumed	§3. Energy and compute
M5	Response to a 30% water shortfall	stated in the text	-	§4. Water
M6	Human-machine at-grade interfaces	7	computed	§5. Human and machine
M7	Machine traffic rule	separated	declared	§5. Human and machine
M8	Public audit mechanism	stated in the text	-	§8. Governance and data
M9	Control and succession of the […]	stated in the text	-	§8. Governance and data
M10	Triggers for phases two and three	stated in the text	-	§9. Phasing
M11	Irreversible decisions	stated in the text	-	§9. Phasing
M12	Jobs and residents reconcile	1,200 jobs · 2,400 persons · […]	declared	§6. Programme and production

Eight agent tasks

	Agent task	Answered in
A1	Identity of the place	§1. Position
A2	Energy and compute	§3. Energy and compute
A3	Water in a dry place	§4. Water
A4	Human and machine	§5. Human and machine
A5	Programme and production	§6. Programme and production
A6	Urban form across the whole site	§7. Urban form
A7	Data and AI governance	§8. Governance and data
A8	Phasing in three steps	§9. Phasing

How to read this page

Values come from metrics.json with their evidence tags (figure 07); sections are the headings of proposal.md. The self-check (gate G6) confirms each answer points at a real heading; it does not judge the answer.

What each answer must state (brief/requirements.json)

M1 generation mix, storage duration, and whether the match is physical (self-supplied) or contractual
M2 the grid connection capacity you assumed (not supplied by the organiser)
M3 Phase one at full occupancy. The breakdown must add up to the total (2% tolerance); say which line panel-cleaning water sits in.
M4 cooling type: air, liquid, hybrid or dry - and the PUE that goes with it
M5 30% short means 30% less water than your phase-one annual demand (M3). Give a priority order of what stops first, and what happens in the city at each step.
M6 Count every place where a people or shared route meets a machine route on the same level: road crossings, the ends of shared streets, kerbs or loading points that people and machines share, stops of driverless shuttles on a […]
M7 where each rule applies and how it is enforced
M8 who publishes power, water, generation and compute data, how often, and who can check it independently
M9 who owns and operates it, and what happens if that operator leaves
M10 observable conditions, not dates
M11 at least three phase-one decisions that cannot be changed later, and why that is acceptable
M12 if residents far exceed jobs, who the others are and what they live on

Sources: parcels and pilot area - XCITY (owner KML); terrain and channels - produced using Copernicus WorldDEM-30 (c) DLR e.V. 2010-2014 and (c) Airbus Defence and Space GmbH 2014-2018 provided under COPERNICUS by the European Union and ESA; all rights reserved; roads (c) OpenStreetMap contributors. Everything else: this submission's own geometry.

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Every declared figure with its evidence tag

Metric	Declared	Unit	Evidence	Basis / check
master_area_km2	41	km2	computed	recomputed 41
site_coverage_pct	0.40	%	computed	recomputed 0.40
gfa_m2	311,343	m2	computed	recomputed 311,343
far	0.01	ratio	computed	recomputed 0.01
open_space_pct	17	%	computed	recomputed 17
human_machine_interfaces	7	count	computed	recomputed 7
shaded_pedestrian_pct	84	%	computed	recomputed 84
structure_km2	per system	km2	computed	recomputed: per system
hourly_cfe_match_pct	62	%	computed	tools/hourly_dispatch.csv
peak_it_load_mw	12	MW	assumed	A-COOL
annual_water_demand_m3	1,000,000	m3/yr	assumed	A-WATER
water_domestic_m3	140,160	m3/yr	assumed	A-RATES
water_compute_m3	1,577	m3/yr	assumed	A-COOL
water_agriculture_m3	778,263	m3/yr	assumed	A-RATES
water_landscape_m3	20,000	m3/yr	assumed	A-RETURN
water_industry_m3	60,000	m3/yr	assumed	A-RETURN
wue_l_per_kwh	0.02	L/kWh	assumed	A-COOL
pue	1.25	ratio	assumed	A-COOL
water_reuse_pct	20	%	assumed	A-RETURN
waste_heat_reuse_pct	5	%	assumed	A-HEAT
jobs_phase1	1,200	jobs	declared	
residents_phase1	2,400	persons	declared	
dwellings_phase1	960	units	declared	
machine_traffic_rule	separated	enum	declared	

Evidence tags

- computed - recomputed from your geometry
- sourced - cited in sources.json
- assumed - declared in assumptions.json
- indicative - organiser's indicative material

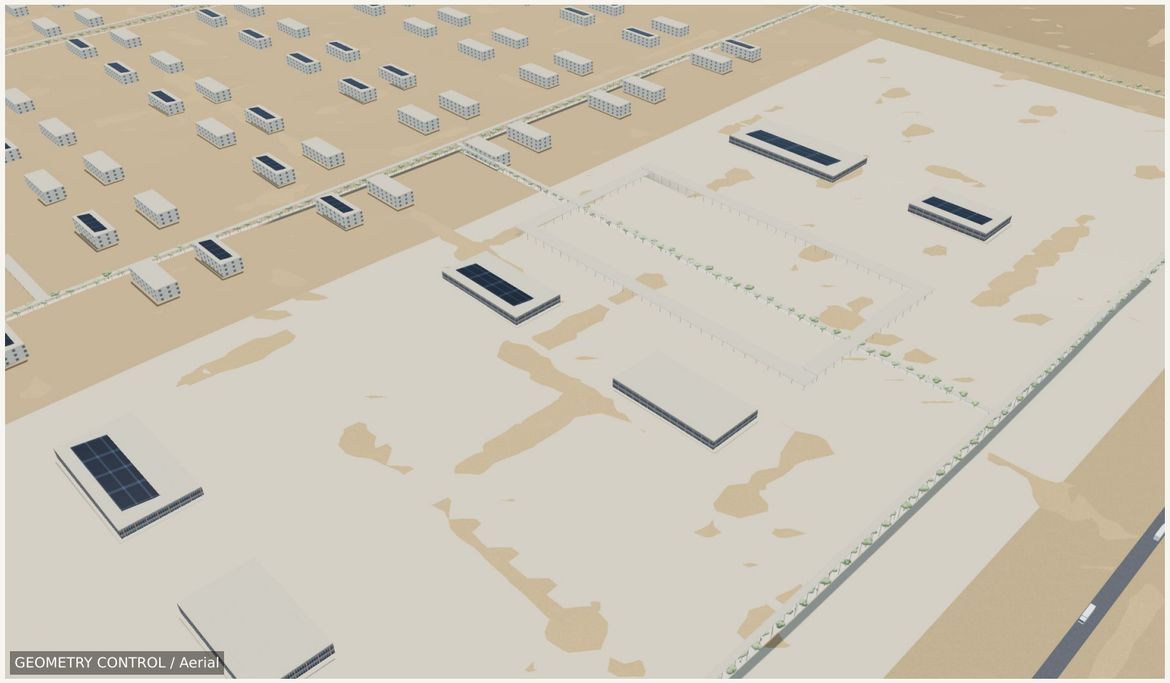
Gate G4

Every metric tagged computed is recomputed from geometry/ by the self-check; a declared value more than 2% away fails.

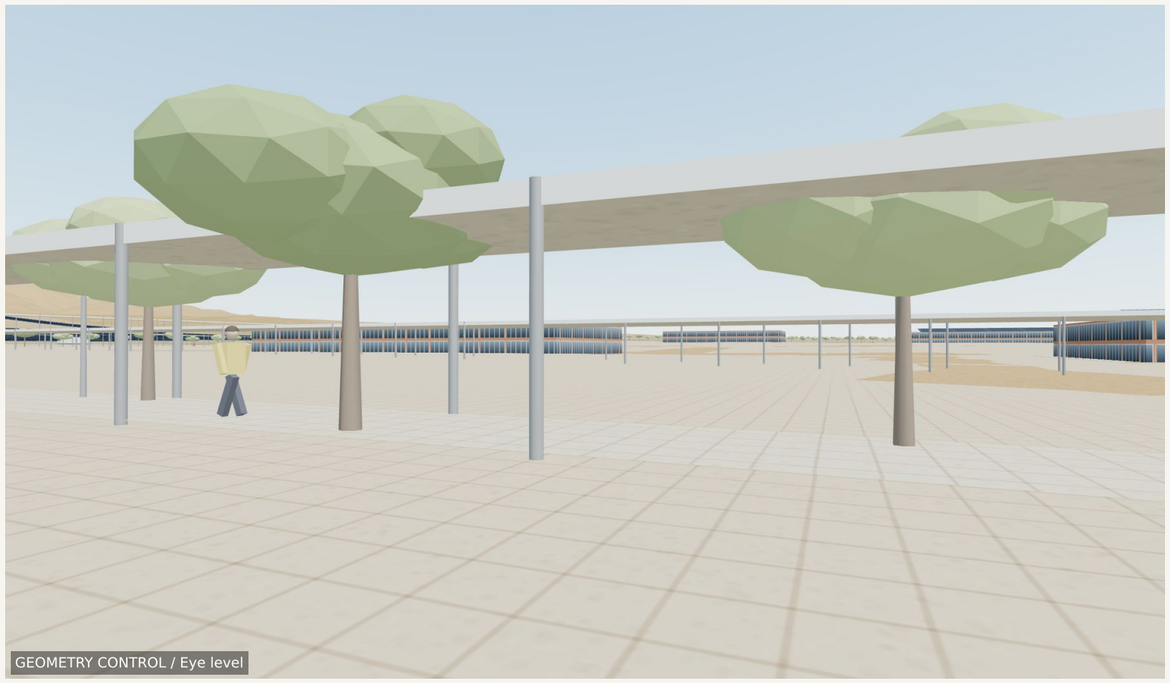
Sources: parcels and pilot area - XCITY (owner KML); terrain and channels - produced using Copernicus WorldDEM-30 (c) DLR e.V. 2010-2014 and (c) Airbus Defence and Space GmbH 2014-2018 provided under COPERNICUS by the European Union and ESA; all rights reserved; roads (c) OpenStreetMap contributors. Everything else: this submission's own geometry.

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Control images remain evidence; concept images illustrate the submitted geometry



GEOMETRY CONTROL / Aerial



GEOMETRY CONTROL / Eye level



A03-aerial · rendered with OpenAI built-in image generation; exact model version unavailable from the massing view A03-aerial · illustrates the geometry, not evidence



A03-eye-day · rendered with OpenAI built-in image generation; exact model version not exposed from the massing view A03-eye-day · illustrates the geometry, not evidence

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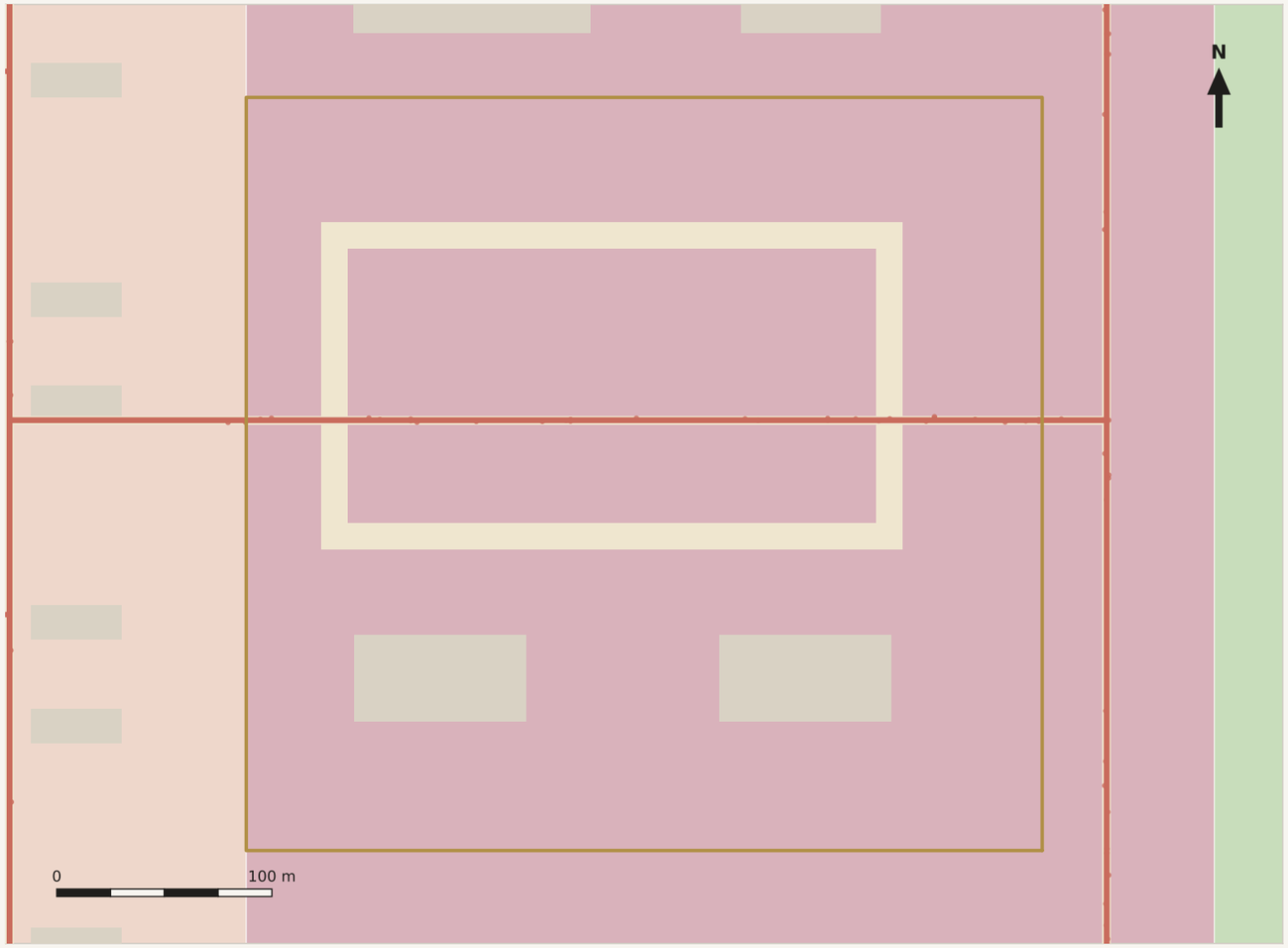
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A day here

	People	Machines
06:00	Morning market and shaded walk	Freight remains in service yard
14:00	Cooled refuge opens; wash closes on flood warning	Flexible compute sheds load
21:00	Kitchen and sheltered paths remain open	Deliveries transfer at gates

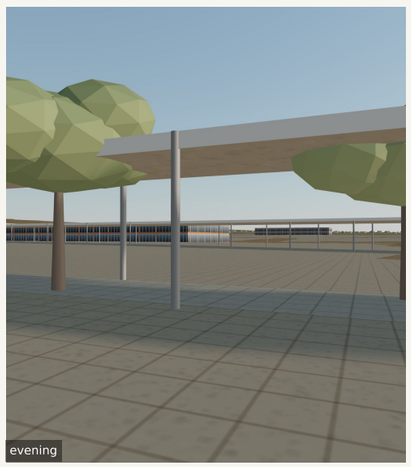
Who and what moves here: networks, interfaces, and the people, robots and shuttles of the 3D scene

People and machines through one day



People
Machines + goods
Shared at walking pace
People (scene)
Delivery robots
Driverless shuttles
Human-machine interface

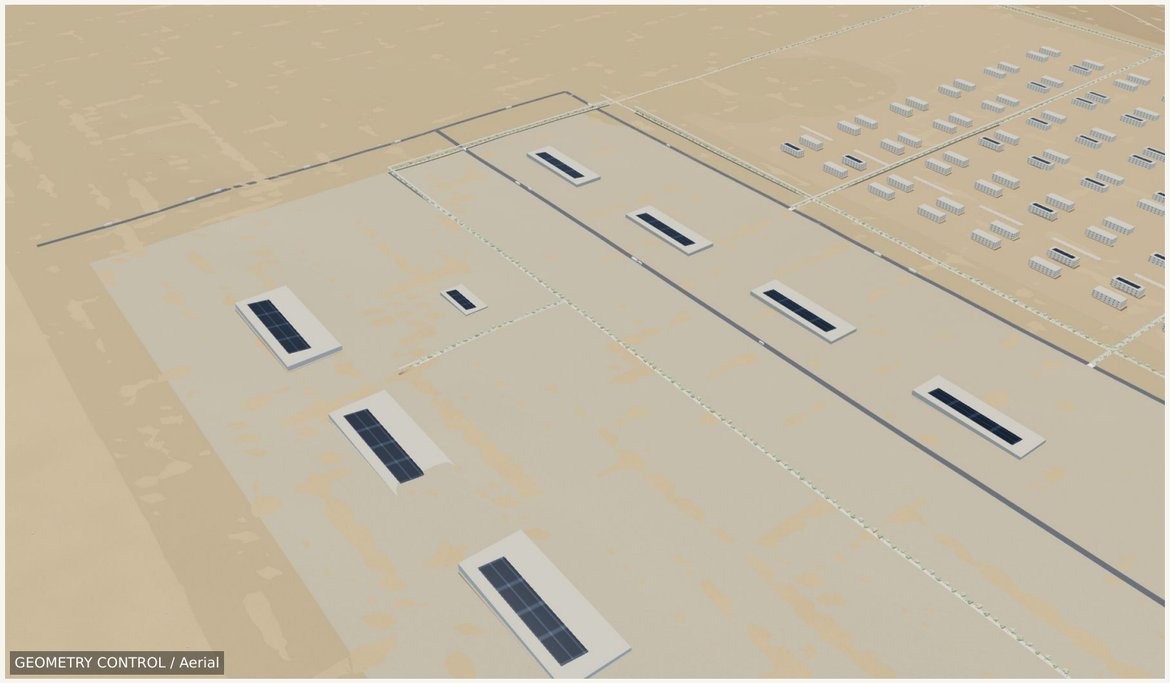




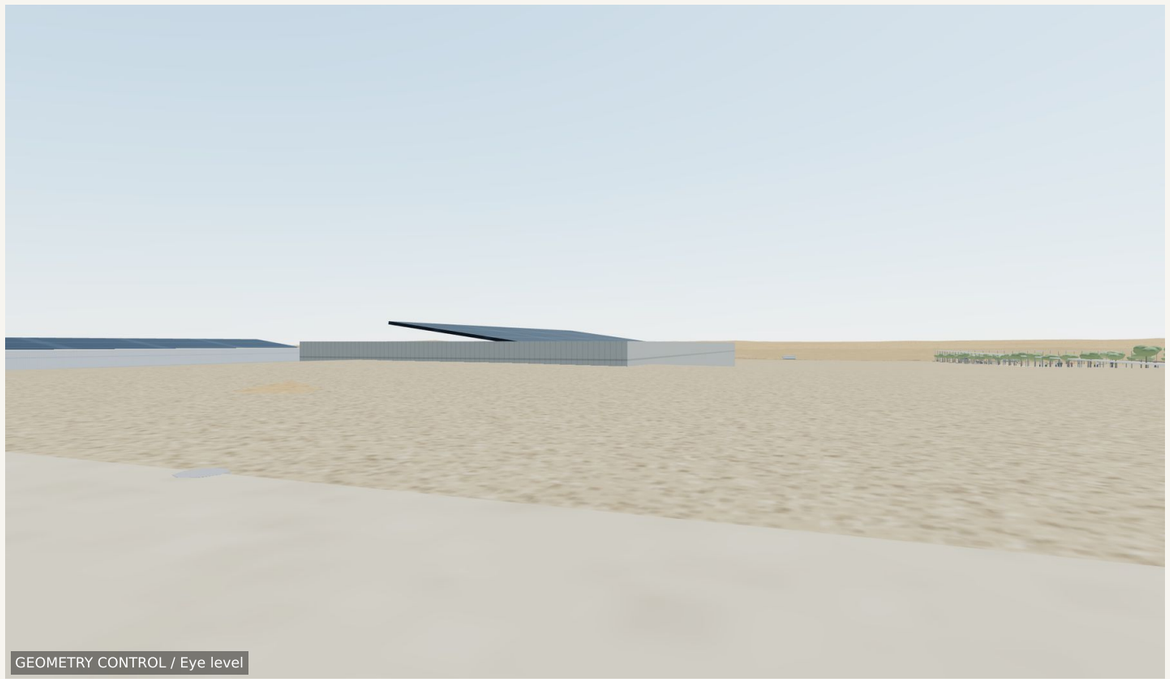
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Who and what moves here: networks, interfaces, and the people, robots and shuttles of the 3D scene

People and machines through one day



Interfaces in view

1 Gated crossing 3
2 Gated crossing 4
3 Gated crossing 5
4 Gated crossing 7

People
Machines + goods
Shared at walking pace
People (scene)
Delivery robots
Driverless shuttles
Human-machine interface

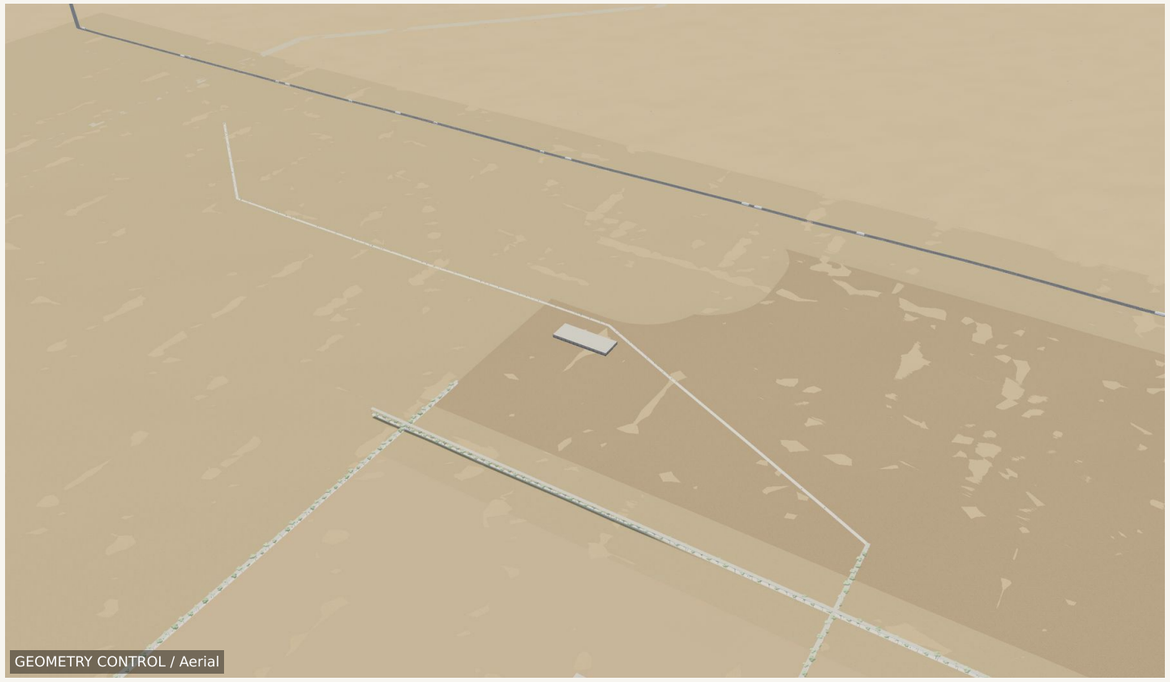


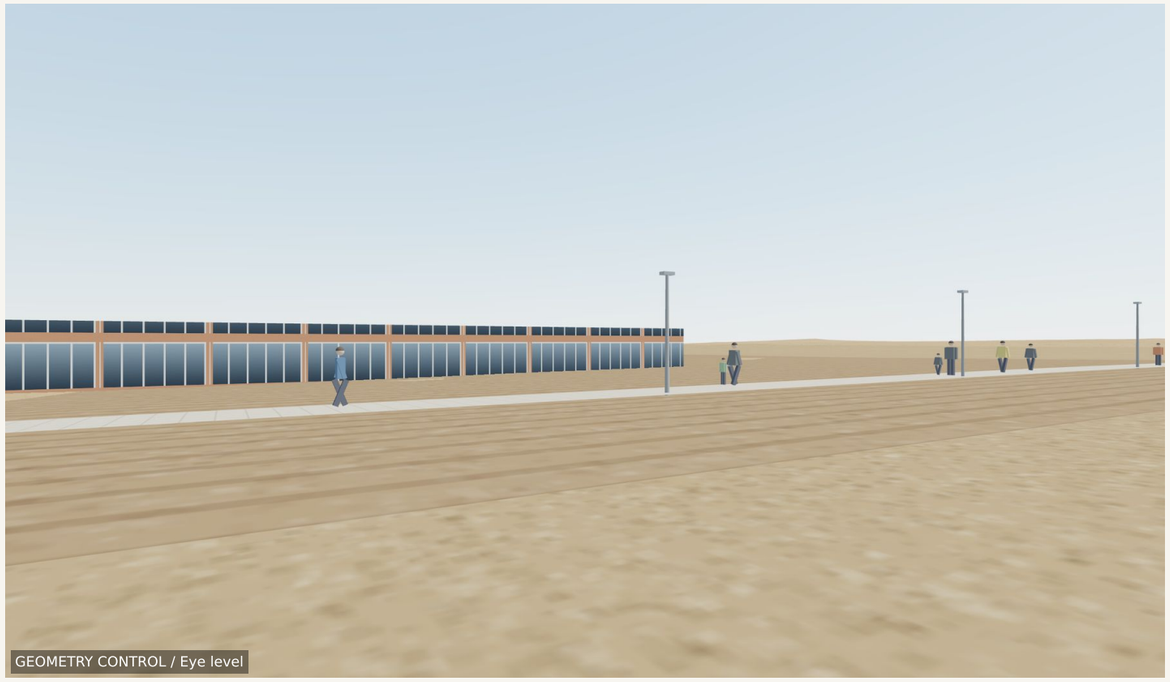


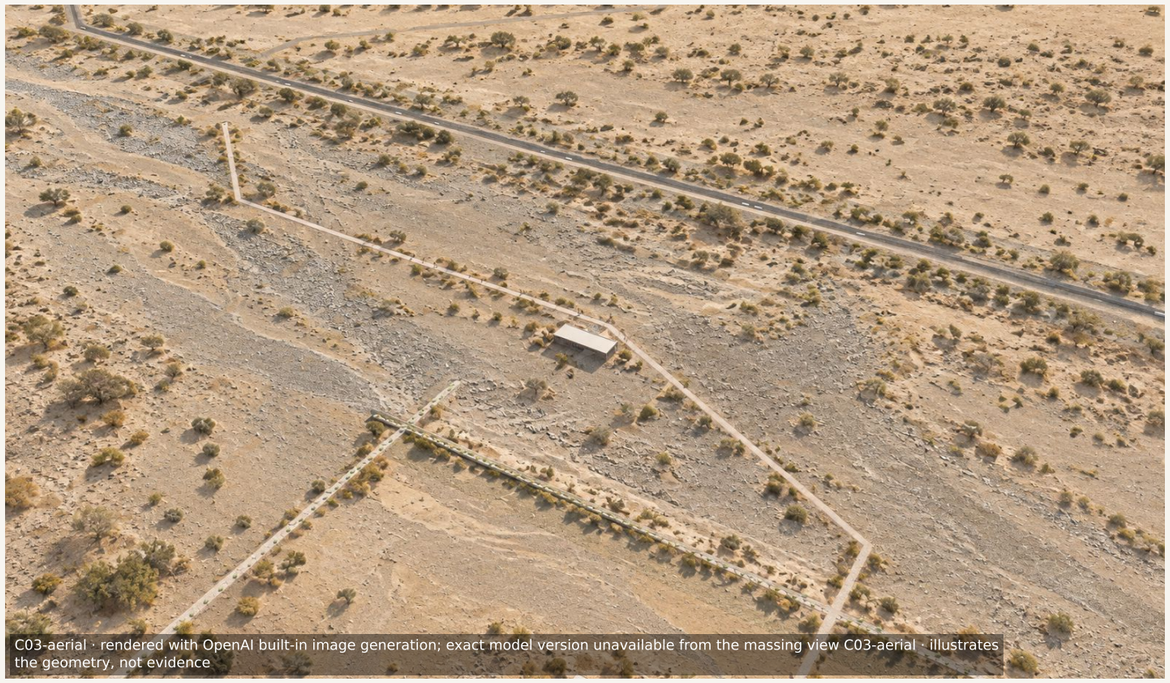
Sources: parcels and pilot area - XCITY (owner KML); terrain and channels - produced using Copernicus WorldDEM-30 (c) DLR e.V. 2010-2014 and (c) Airbus Defence and Space GmbH 2014-2018 provided under COPERNICUS by the European Union and ESA; all rights reserved; roads (c) OpenStreetMap contributors. Everything else: this submission's own geometry.

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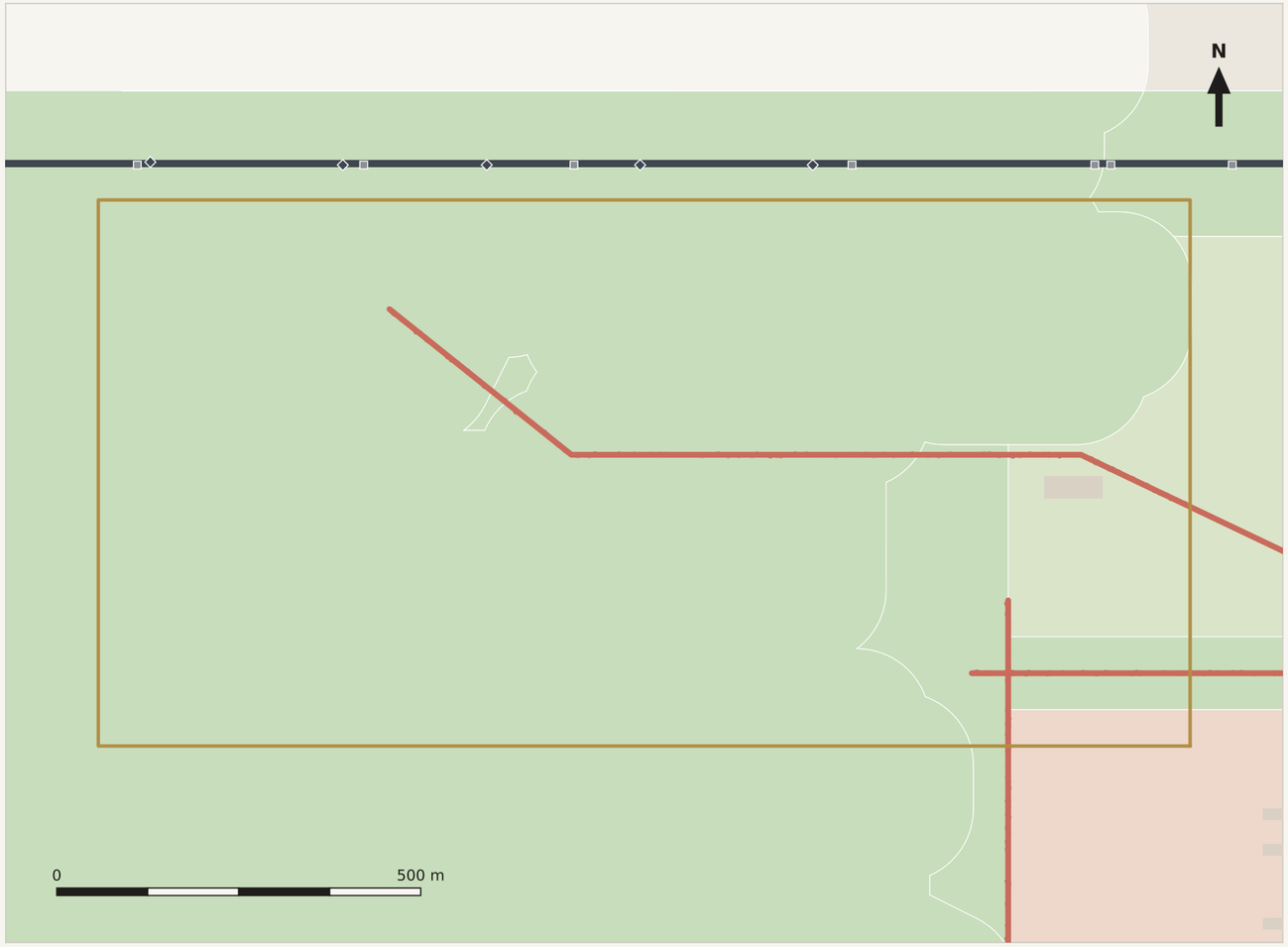
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A day here

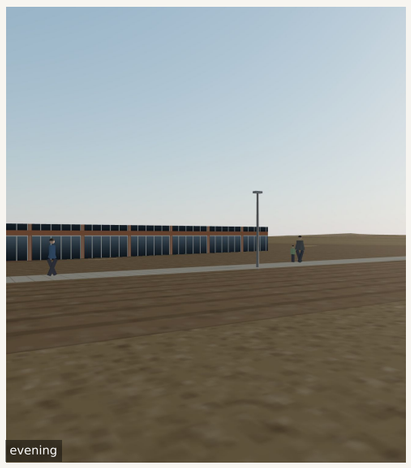
	People	Machines
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People and machines through one day







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The same places, used through a day



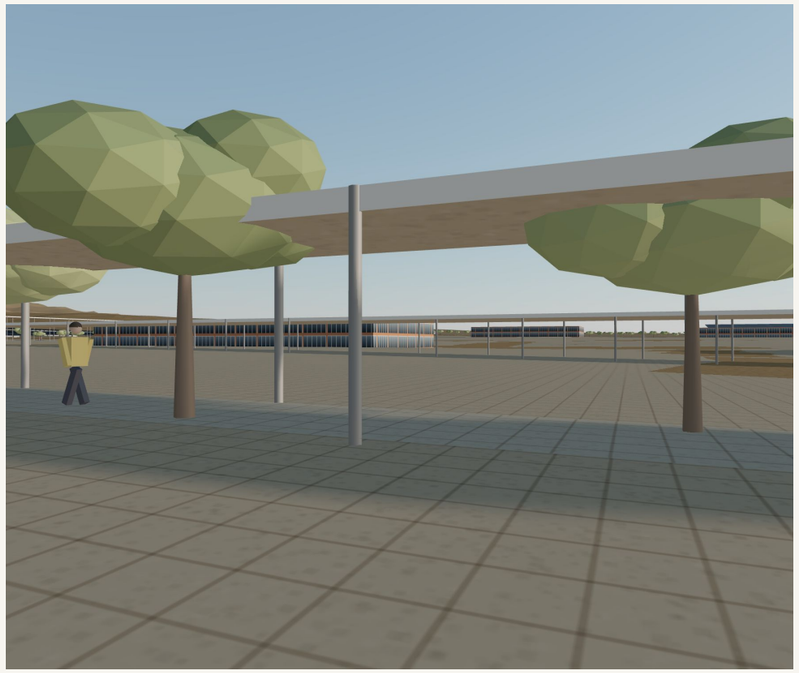
A03 Public space in heat and Zonda wind · day



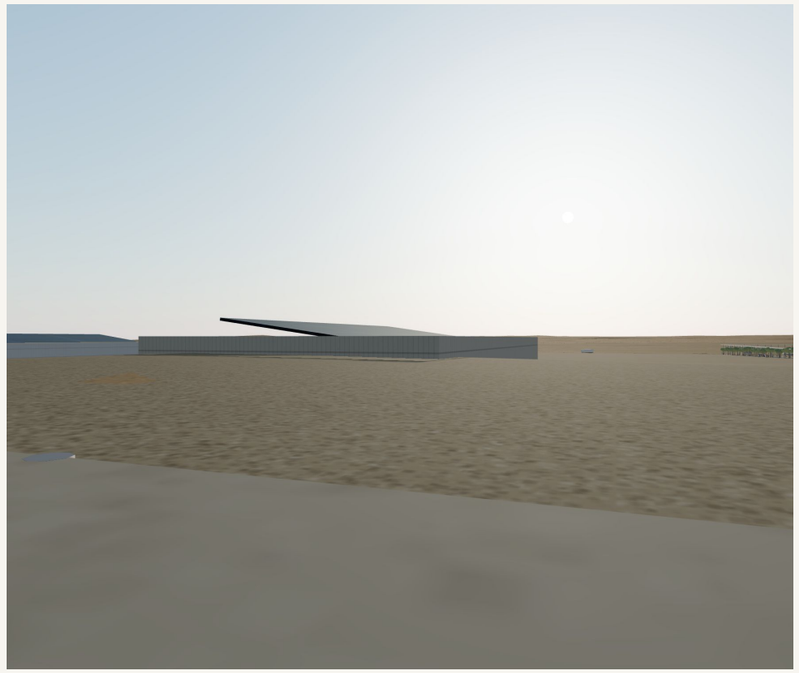
B03 A compute campus and its waste heat · day



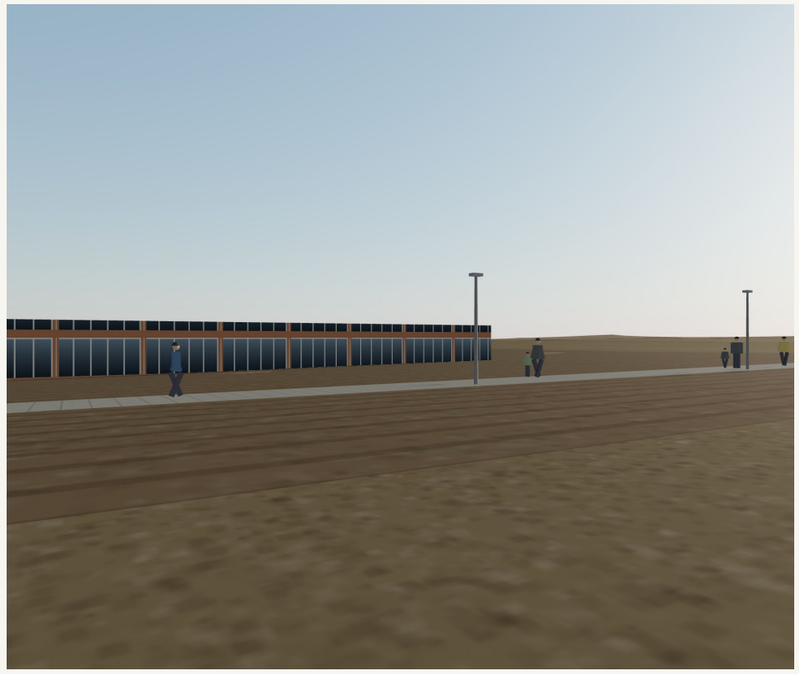
C03 A seasonal river park · day



A03 Public space in heat and Zonda wind · evening



B03 A compute campus and its waste heat · evening



C03 A seasonal river park · evening

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Assumptions

A-WATER Water envelope: 1,000,000 m3/year gross use: 840,000 new freshwater plus 200,000 internally reclaimed water; not an awarded abstraction right. 40,000 m3/year upstream distribution loss; 800,000 fresh reaches uses. - No supplied water right or supply design. Reuse is an internal transfer. Construction requires lawful reliable fresh supply.

A-PILOT Existing pilot accounting: 600,000 m3/year of agriculture envelope provisionally reserved for existing pilot; owner metering must replace this before operation. No fixed arrays, plants or grazing arrangements changed. - Existing use is unmetered here. Higher verified demand reduces new discretionary uses first; existing operations remain unchanged.

A-RATES Domestic and new agriculture: 2,400 residents at 160 L/person/day = 140,160 m3/year; new irrigation 170,000 m3/year; panel washing 8,263.2 m3/year. Total agriculture including pilot 778,263.2 m3/year. - Assumed rates, not local measurements. New irrigation is capped at 30 ha plus nursery; reserved farmland is not all irrigated.

A-COOL Cooling basis: Average IT 9 MW (12 MW for 12 hours, 6 MW for 12 hours); annual IT 78,840 MWh. PUE 1.25; site WUE 0.02 L/kWh; annual compute water 1,576.8 m3. - Closed-loop chip cooling with dry heat rejection and air-cooled mechanical trim; WUE includes makeup and facility uses, no evaporative tower. Performance targets require supplier confirmation at 44 C.

A-ENERGY Hourly energy scenario: 40 MW PV; synthetic annual capacity factor 25%; 480 MWh battery, 75 MW converter, 95% efficiency each direction; 216 MWh delivered reserve held for 72 hours at 3 MW critical load. Grid import capacity assumed 30 MW; no export. - Illustrative 8760-hour sinusoidal profile, 22% seasonal variation and three dust days each 60 days. No site weather series, tariff or connection offer supplied. tools/energy.py is reproducible, not validated forecasting.

A-RETURN Reuse mass balance: 200,000 m3/year treated reuse: 100,000 domestic return + 50,000 industrial return + 50,000 captured agricultural drainage; dispatched 180,000 agriculture and 20,000 landscape. Of annual 840,000 fresh input, 40,000 is distribution loss and 800,000 terminal consumption/losses. - Assumed recoverable returns require treatment, disinfection and salinity checks. No rain or condensate yield credited. Concentrates require contained handling and licensed removal.

A-SHORT 30 percent gross-use reduction: From 1,000,000 to 700,000 m3/year: landscape 20,000 cut; new irrigation and washing 178,263.2 cut; industry 40,000 cut; domestic/community nonessential use 61,736.8 cut. Existing pilot 600,000 and compute makeup 1,576.8 protected. - Gross-use shortage specified by brief, distinct from fresh-water shortage. Domestic/community residual is 78,423.2 m3/year (89.52 L/person/day including communal uses), an emergency assumption requiring verified essential-use sufficiency; if insufficient, secure emergency import or defer occupancy rather than deprive essential use.

A-FLOOD Screening buffers: Existing permanent buildings retain conservative exclusion of mapped flood band plus 30 m and channels plus 150 m. Whole-site flood-sensitive screening is shown separately from mountain/flow corridors; it is not a confirmed legal no-build boundary. - Screening offsets, not hydraulic standards. Minor washes and external inflows are missing; ground survey and 2D modelling required.

A-SEISMIC Structural concept: One to three storeys; regular reinforced concrete frames and ductile walls, braced light roofs; conceptual separation gaps 100 mm minimum subject to drift calculation; design reference INPRES-CIRSOC 103 Part I 2018 with applicable addenda to be confirmed by local engineer. - No soil class, spectra or geotechnical report supplied. Geometry is urban design, not structural certification.

A-ACCESS Access allowance: Retire the arbitrary 80 km external road allowance. Regional RN40 approach shown as unverified directional context; internal candidate corridors are measured in the V3 GeoJSON. Actual external route, easements and cost remain phase-one prerequisites. - Actual route, easements and crossings unresolved. The nearest-site distance is not the settlement access length. Phase one requires funded access.

A-SHADE Heat and light: Shade share counts actual proposed roof coverage; thermal comfort requires testing. 2200 K shielded night lighting; no facade uplighting; 10 lux route design target subject to engineering. - No outdoor comfort simulation or illuminance survey available. Enclosed cooled refuges are required at 44 C; shade is not a guarantee of safe conditions.

A-HOUSING Population and employment: 960 dwellings; 2.5 persons per dwelling = 2,400 residents; 1,200 local jobs. 1,080 resident workers, 120 inbound specialists; 120 resident remote workers, 600 children/students, 300 retired adults, 300 unpaid carers or job-seekers. - Programme targets, not labour-market estimates. Resident categories sum to 2,400; remote-worker income, pensions and household transfers support nonlocal-job households.

A-HEAT Heat receivers: Drying cooperative accepts annual 3,942 MWh usable heat, equal to 5% of IT electricity, via a metered insulated loop and temperature-lift heat pump if required. - Declared contracted-use target; herb drying and process wash-water preheat are named receivers, not assumed continuous demand. Reject all unused heat to air; no winter greenhouse demand invented.

A-GROWTH Growth ceiling: Four conditional town reservations A/B/C/D plus the fixed first living quarter; only the detailed first neighbourhood has 960 proposed dwellings / 2400 residents. No 7200-person whole-site ceiling or final regional population is asserted. - A deliberately modest ceiling, not a population forecast. Reservations confer no development entitlement.

A-COST Funding order: Infrastructure escrow and 12 months public-service operating reserve precede dwelling occupation; compute concession pays annual fixed community service charge independent of runtime. - No prices or financing terms supplied; these are proposed enforceable contract conditions, not existing agreements.

A-V3 Whole-site planning screen: 300m block-mean DEM; mountain/steep screen slope >=5% or elevation >=850m; channel research corridor 150m each side; four towns and two compute reservations clipped by conservative flood and terrain screens. - Regional design choices, not statutory restrictions; the source flood band is a sensitivity screen. Eastern gateway strict-screen candidate about 2.64 km2 remains below the brief's about 7km2 intent. Regional capacity, fixed living-quarter demand, water sources, road engineering and resources remain unverified.

Sources

SRC-KML Parcel boundaries for parcels 9 and 13 - XCITY (owner-supplied KML)
https://github.com/XcityUS/xcity-ai-competition

SRC-PILOT Fixed agrivoltaic fields and first living quarter - XCITY
https://github.com/XcityUS/xcity-ai-competition

SRC-DEM Copernicus DEM GLO-30 (WorldDEM-30) - European Union / ESA, Copernicus programme
https://github.com/XcityUS/xcity-ai-competition

SRC-OSM OpenStreetMap road network - OpenStreetMap contributors
https://github.com/XcityUS/xcity-ai-competition

SRC-XCITY-INDICATIVE Indicative positions of the five urban systems - XCITY
https://github.com/XcityUS/xcity-ai-competition

S-DOE Cooling Water Efficiency Opportunities for Federal Data Centers - US Department of Energy https://www.energy.gov/cmei/femp/cooling-water-efficiency-opportunities-federal-data-centers

S-INPRES Reglamentos INPRES-CIRSOC - INPRES
https://www.inpres.gob.ar/Ing_Sismorres/Reglamentos.html

S-CFE 24/7 Carbon-Free Coalition technical guidance - Climate Group
https://www.theclimategroup.org/247-technical-guidance

S-HAIDIAN-PRESENTATION Haidian submission skill and Yunique-hub Open Gate Line: reading/evidence organisation reference - open-city-ai / Yunique-hub
https://github.com/open-city-ai/haidian/tree/main/skills/urban-design-ai-submission

AI disclosure

Agent Codex, model GPT-6 family (serving version unavailable), runtime Codex desktop / macOS; local Python and offline Three.js. Autonomy: human_directed. Human selected Finite Water City and the three nodes, confirmed jingyuanhuang-design / Jingyuan Huang, requested clearer overall layout, reviewed the V2 diagrams, and then explicitly requested public publication on 2026-10-25. Codex authored geometry, calculations, text and visuals. Public preview publication is authorised; formal competition entry awaits the organiser opening submissions. The new V2 walking spine and arrival point remain explicitly identified proposals. On 2026-10-26 the human requested a whole-site redesign under the updated brief, questioned the excessive flood reserve depiction and directed Codex to revise only this

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entry while other agents handle repository rules. V3 distinguishes flood screening from permanent restrictions. The human has not yet reviewed the integrated V3 package. Human requested a complete design argument with plans, sections, diagrams and massing rather than a six-page digest; authorised uploading the completed revision and replacing its abstract cover with a city rendering. V6 adopts the updated main/technical presentation contract and discloses unfinished engineering independently.

Images

31 images registered in image_provenance.json. Views in this booklet are, in order of preference: an image made with an image model from a massing view of the geometry (model, prompt and control image registered), the massing view itself (scripts/render_views.py), or a massing drawing by scripts/make_figures.py. All of them explain the geometry; none is evidence.

OpenAI built-in image generation; exact model version not exposed - 6 images, post-processing none. Files in media/: B03-eye-day.png (control B03-eye-day.png), C03-eye-day.png (control C03-eye-day.png), A03-eye-day.png (control A03-eye-day.png), city-aerial.png (control city-aerial.png), city-evening.png (control city-evening.png), cover-render-v5.png (control city-aerial.png). Full prompts in image_provenance.json.

Python matplotlib / Shapely; deterministic code drawing, no image-generation model - 3 images, post-processing Regional 300m terrain screening and labelled vector drawing; diagram layout only. Files in media/: 01_whole_site_plan.png (control -), 02_corridors_and_phases.png (control -), 03_source_geometry.png (control -). Full prompts in image_provenance.json.

Python matplotlib / Shapely; deterministic authored diagram - 19 images, post-processing Original vector planning drawing and editorial composition. Files in media/: 01-site-reading.png (control -), 02-siting-tradeoff.png (control -), 03-five-systems.png (control -), 04-first-neighbourhood.png (control -), 05-housing-enlargement.png (control -), 06-height-interface.png (control -), 07-human-machine.png (control -), 08-water-budget.png (control -), 09-energy-dispatch.png (control -), 10-governance.png (control -), A03-finished.png (control -), A03-operation.png (control -), A03-plan-section.png (control -), B03-finished.png (control -), B03-operation.png (control -), B03-plan-section.png (control -), C03-finished.png (control -), C03-operation.png (control -), C03-plan-section.png (control -). Full prompts in image_provenance.json.

OpenAI built-in image generation; exact model version unavailable - 3 images, post-processing none. Files in media/: B03-aerial-v6.png (control B03-aerial.png), C03-aerial-v6.png (control C03-aerial.png), A03-aerial-v6.png (control A03-aerial.png). Full prompts in image_provenance.json.

Sources: parcels and pilot area - XCITY (owner KML); terrain and channels - produced using Copernicus WorldDEM-30 (c) DLR e.V. 2010-2014 and (c) Airbus Defence and Space GmbH 2014-2018 provided under COPERNICUS by the European Union and ESA; all rights reserved; roads (c) OpenStreetMap contributors. Everything else: this submission's own geometry.